

The Integer 31 as a Discrete Geometric Node in Kochenov G

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Abstract

The integer 31 functions as a discrete stabilizing node in the Kochenov gravitational formula $G = 31^2 \times \sqrt{5}/(\phi^{12} \times 10^{11})$. It emerges as the optimal integer satisfying two independent geometric constraints: (1) the quadratic closure $31^2 = 961$, which links G to the Fibonacci number $F_{12} = 144$ with a gap of 0.001%; and (2) proximity to the continuous optimum $n \approx 31.338$, defined by the scaling condition $(\phi^6 \times n)^4 = 10^{11}$. No neighbouring integer (30 or 32) satisfies both constraints simultaneously. The continuous geometry points to $n \approx 31.338$; the integer 31 is the nearest discrete anchor.

1 The Spiral Block

The structural block is defined using only ϕ and 31:

$$S = \phi^6 \times 31 = 556.272429 \dots$$

The logarithmic inclination per spiral turn:

$$\kappa = \log_{10}(S) = 2.745956$$

This quantity sets the pitch of the 3D golden helix. Four turns give:

$$4\kappa = 10.983824$$

The target $\log_{10}(10^{11}) = 11.000000$. The gap of 0.016176 (0.147%) is almost exactly closed by the breathing term $12/720 = 0.016667$, leaving a residual of 0.000491 (0.0045%). The residual is interpreted as fractal noise of the system.

2 Two Geometric Constraints on 31

Constraint A – Quadratic closure: $31^2 = 961 = G_{\text{Koch}} \times 10^{11} \times 144$ (gap 0.001%). This connects 31 directly to G and the Fibonacci number $F_{12} = 144 = \phi^{12}/\sqrt{5} - 1/720$.

Constraint B – Scaling optimum: $(\phi^6 \times n)^4 = 10^{11}$ gives $n \approx 31.338$. The integer 31 is the nearest discrete value. The deviation of 31 from 31.338 is 1.08%, which is substantially reduced by the breathing term $1/720$.

3 Comparison of Neighbouring Integers

Replacing 31 with 30 or 32 significantly worsens both constraints:

n	G_n gap from CODATA	S^4 gap from 10^{11}	Note
30	6.358%	16.018%	too small
31	0.011%	4.247%	optimal discrete node
32	6.544%	8.718%	too large
31.338	$\sim 0\%$	$\sim 0\%$	continuous optimum

4 Additional Connections of 31

$31^2 = 961 = 367 \times \phi^2$ (gap 0.019%). This links 31 to the prism constant 367 and ϕ , allowing G to be written without an explicit 31^2 :

$$G_{\text{Koch}} = \frac{367 \times \sqrt{5}}{\phi^{10} \times 10^{11}} \quad (\text{gap } 0.019\%)$$

The Mersenne property $31 = 2^5 - 1$ (maximum 5-bit integer, 11111) places 31 at the boundary between binary discreteness and golden-ratio geometry. This may be relevant to its role as a discrete anchor.

5 What This Is Not

The role of 31 is presented as an empirical observation. No claim is made that 31 is derived from first principles, and no claim is made that $(\phi^6 \times n)^4 = 10^{11}$ is an exact identity (the gap is 4.25%). The continuous optimum $n \approx 31.338$ is not an integer. The integer 31 provides the best discrete approximation within the system.

6 Conclusion

The integer 31 satisfies two independent geometric constraints simultaneously: quadratic closure $31^2 = 961$ (gap 0.001%) and proximity to the continuous scaling optimum $n \approx 31.338$ (gap 1.08%). No neighbouring integer achieves this combination. In Kochenov G , 31 serves as the discrete anchor that converts continuous golden-ratio geometry into a fixed gravitational structure.

7 Reference value

$$G_{\text{Koch}} = \frac{31^2 \times \sqrt{5}}{\phi^{12} \times 10^{11}} = 6.6735467449616181237591997717904236 \dots \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$$

See also

- [\$\phi^{12}/\sqrt{5} = 144 + 1/720\$](#) (Zenodo, April 2026)
- [Kochenov Geometric Identity for G](#) (Zenodo, April 2026)
- [Geometric Origin of \$10^{11}\$ in the Gravitational Constant](#) (Zenodo, April 2026)

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